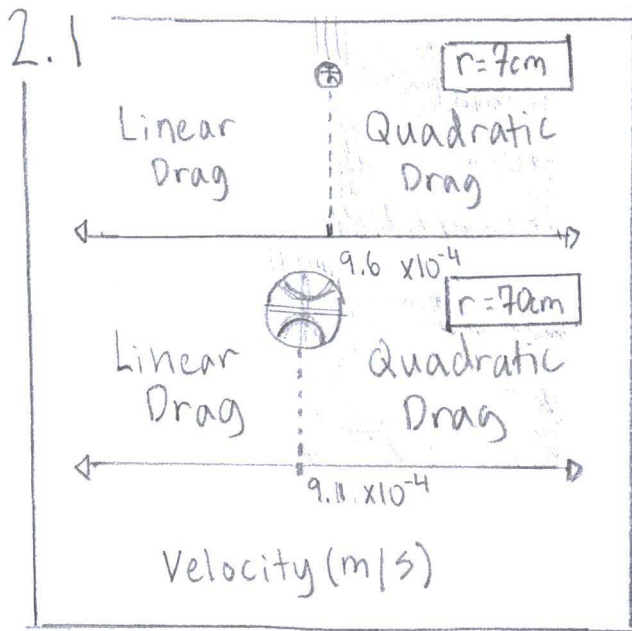


Chapter 2: Projectiles and Charged Particles:



Diameter - 70cm:

(2.7) "Ratios between drag"

$$\frac{f_{\text{quad}}}{f_{\text{lin}}} = \frac{cv^2}{bv} = \frac{\gamma D}{\beta} v$$

when for spheres

$$\gamma = 0.25 \text{ N}\cdot\text{s}^2/\text{m}^4$$

$$\beta = 1.6 \times 10^{-4} \text{ N}\cdot\text{s}/\text{m}^2$$

D = Diameter

v = velocity

Linear equals Quadratic drag at:

$$\frac{f_{\text{quad}}}{f_{\text{lin}}} = \frac{\gamma D}{\beta} v$$

$$= 1$$

$$v = \frac{\beta}{\gamma D}$$

$$= \frac{1.6 \times 10^{-4} \text{ N}\cdot\text{s}/\text{m}^2}{0.25 \text{ N}\cdot\text{s}^2/\text{m}^4 \cdot 0.07 \text{ m}}$$

$$= 9.6 \times 10^{-4} \text{ m/s}$$

$$= 9.6 \times 10^{-4} \text{ m/s}$$

Diameter - 70cm:

Linear equals Quadratic drag at:

$$\frac{f_{\text{quad}}}{f_{\text{lin}}} = \frac{\gamma D}{\beta} v$$

$$= 1$$

$$v = \frac{\beta}{\gamma D}$$

$$= \frac{1.6 \times 10^{-4} \text{ N} \cdot \text{s/m}^2}{0.25 \text{ N} \cdot \text{s}^2/\text{m}^4 \cdot 0.70 \text{ m}}$$

$$= 9.1 \times 10^{-4} \text{ m/s}$$

Both baseballs are nearly quadratic drag (f_{drag}) from the initial instant in freefall.

2.2

(2.82) "Stokes Law"

$$f_{\text{lin}} = 3\pi\eta Dv$$

The equation above is expression 2.3, with $b = \beta D$.

(2.3) "Linear Drag"

$$f_{lin} = b \cdot v$$

$$= \beta \cdot D \cdot v$$

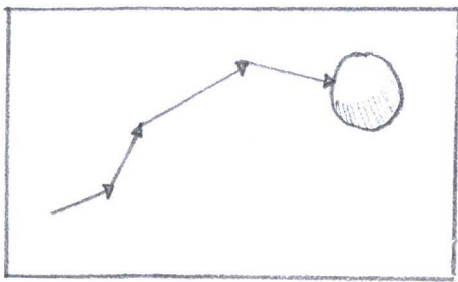
$$= 1.6 \times 10^{-4} \text{ N} \cdot \text{s/m}^2 \cdot D \cdot v$$

$$f_{lin, \text{stokes}} = 3\pi \eta D \cdot v$$

$$= 3\pi \cdot 1.7 \times 10^{-5} \text{ N} \cdot \text{s/m}^2 \cdot D \cdot v$$

$$= 1.6 \times 10^{-4} \text{ N} \cdot \text{s/m}^2 \cdot D \cdot v$$

2.3



"moving sphere
in a fluid"

a) (2.82) "Stokes Law for spheres"

$$f_{lin} = 3\pi \eta \cdot D \cdot v$$

(2.84) "Stokes Law for spheres"

$$f_{quad} = k \rho A v^2 \quad \text{where } k \leq 1$$

ρ = "Density
of fluid"

A = "Cross-
sectional
Area"

η = "Viscosity"

$$\frac{f_{quad}}{f_{lin}} = \frac{k \rho A v^2}{3\pi \eta D v}$$

$$= \frac{Dvp}{\eta} \cdot K \cdot \frac{1}{12}$$

$$= R \cdot \frac{1}{48}$$

Useful
Relations:

$$A = \pi r^2$$

$$D = 2 \cdot r$$

$$K \sim 1/4$$

b) Steel ball bearing:

$$D = 2 \text{ mm}$$

$$v = 5 \text{ cm/s}$$

Fluid - Glycerin:

$$\rho = 1.3 \text{ g/cm}^3$$

$$\eta = 12 \text{ N}\cdot\text{s/m}^2 \text{ @ STP}$$

Radius: $R = \frac{Dvp}{\eta}$

$$= \frac{2 \text{ mm} \cdot 5 \text{ cm/s} \cdot 1.3 \text{ g/cm}^3 \cdot \frac{1 \text{ m}}{100 \text{ cm}} \cdot \frac{1 \text{ kg}}{1000 \text{ g}} \cdot \frac{1000}{1 \text{ m}}}{12 \text{ N}\cdot\text{s/m}^2 \cdot \left(\frac{10^3 \text{ mm}}{1 \text{ m}} \right)}$$

$$= 0.13 \text{ m}$$

$$\sim 0.1 \text{ m}$$

2.4

$$\begin{aligned} \text{a) Rate} &= A [\text{m}^2] \cdot v [\text{m/s}] \cdot \rho [\text{g/cm}^3] \\ &= [\text{g/s}] \end{aligned}$$

$$b) \text{ Net Force} = A [m^2] \cdot v^2 [(m/s)^2] \cdot \rho [g/cm^3] \\ = \left[\frac{kg \cdot m}{s^2} \right]$$

c) (2.84) "Stokes quadratic drag"

$$F_{\text{quad}} = K \rho A v^2$$

(2.3) "Quadratic Drag"

$$F_{\text{quad}} = C v^2 \\ = \gamma D^2 v^2 \quad \dots \text{when } C = \gamma D^2$$

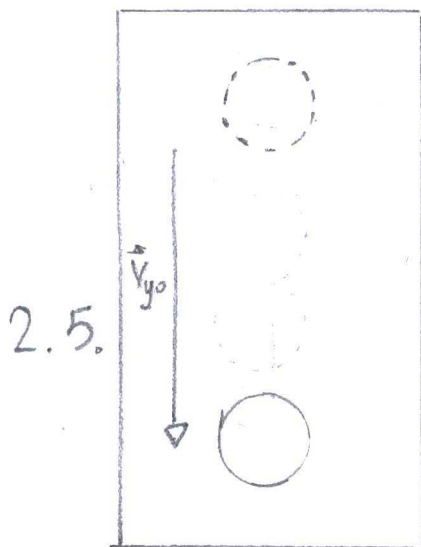
Line 2.84 equates to 2.3

When $A = \pi \cdot D^2 / 4$, a sphere's cross-section.

$$F_{\text{quad}} = \frac{K \rho \pi D^2 v^2}{4}$$

$$= \left(\frac{K \rho \pi}{4} \right) D^2 v^2$$

$$= \gamma \cdot D^2 v^2 \quad \dots \text{with } K = 1/4 \\ \rho = 1.29 \text{ kg/m}^3$$



"Projectile...thrown vertically down."

A projectile velocity ($\vec{v}_{y0}$) above terminal velocity ($\vec{v}_{ter}$) is initially possible, but not throughout the freefall.

(2.26) "Terminal velocity"

$$\vec{v}_{ter} = \frac{mg}{b} \quad [\text{for linear drag}]$$

(2.25) "Vertical motion with linear drag"

$$m\dot{v}_y = mg - bv_y$$

$$= -b(\vec{v}_y - \vec{v}_{ter})$$

$$\int \frac{dv_y}{v_y - v_{ter}} = -\frac{b}{m} \int dt$$

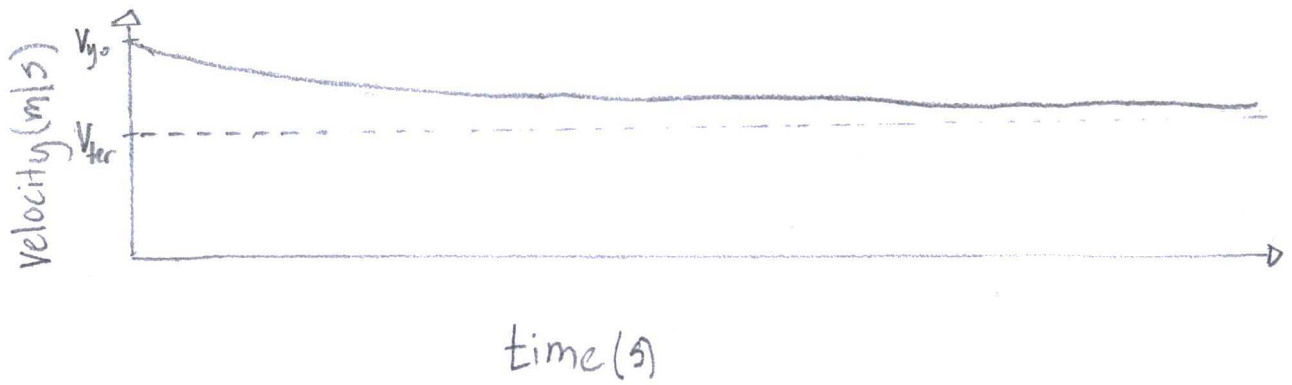
$$\ln(v_y - v_{ter}) = -\frac{b}{m}t + A$$

$$v_y = A e^{-bt/m} + v_{ter}$$

$$@t=0, v_{y0} = A + v_{ter}$$

$$A = v_{y0} - v_{ter}$$

Velocity function:
$$v(t) = (v_{y0} - v_{ter}) e^{-bt/m} + v_{ter}$$



2.6.

a) (2.33) "Projectile dropped from rest"

$$V_y(t) = V_{ter} (1 - e^{-t/\tau})$$

$$= V_{ter} \left(1 - \left(1 + \left(\frac{-t}{\tau} \right) + \frac{\left(\frac{-t}{\tau} \right)^2}{2!} + \dots \right) \right)$$

$$\approx \frac{V_{ter} \cdot t}{\tau}$$

$$\approx g \cdot t$$

Taylor series "Exponential" $e^x = 1 + x + \frac{x^2}{2!} + \dots$

b) (2.35) "Falling object... position"

$$y(t) = \int_0^t V_y(t') dt'$$

$$= V_{ter} \cdot t + (V_{y0} - V_{ter}) \tau (1 - e^{-t/\tau})$$

$$= V_{ter} \cdot t + (V_{y0} - V_{ter}) \tau \left(1 - \left(1 + \left(\frac{-t}{\tau} \right) + \frac{\left(\frac{-t}{\tau} \right)^2}{2!} + \dots \right) \right)$$

$$= V_{ter} \cdot t - V_{ter} \cdot \tau \left(-\frac{t}{\tau} + \frac{t^2}{2\tau^2} + \dots \right)$$

... When $V_{y0} = 0$ from problem

$$= \frac{1}{2} g t^2$$

2.7

(1.17) "Newton's Second Law"

$$\vec{F} = m \cdot \vec{a}$$

$$= m \frac{d\vec{v}}{dt}$$

$$m \int_{\vec{v}_0}^{\vec{v}} \frac{d\vec{v}}{\vec{F}(\vec{v})} = \int dt$$

Time as a function of velocity:

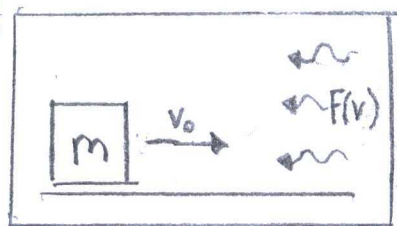
$$\int dt = \frac{m}{F_0} \int_{v_0}^v d\vec{v}$$

$$t = \frac{m}{F_0} (v - v_0)$$

Velocity as a function of time:

$$v = \frac{F_0}{m} t + v_0$$

2.8



"A mass... Coasts
along x-axis...
with 'drag' force"

Velocity in terms of time:

(1.17) "Newton's Second Law"

$$\vec{F} = m \vec{a}$$

$$= m \cdot \frac{d\vec{v}}{dt}$$

$$m \int \frac{dv}{F(v)} = \int dt$$

$$m \int \frac{dv}{(-cv^{3/2})} = \int_0^t dt$$

$$V^{-1/2} = \frac{c}{2m} t + A$$

... Drag force given
from problem 2.8
 $F(v) = -cv^{3/2}$

@ $t=0$, $v=V_0$, so $A = V_0^{-1/2}$

$$V^{-1/2}(t) = \frac{c}{2m} t + V_0^{-1/2}$$

$$V(t) = \frac{4m V_0}{ct\sqrt{V_0} + 2m}$$

Time at rest is infinite when
 $V(t) = 0$,

2.9

(2.2a) "Vertical Force with Drag"

$$m \dot{V}_y = -b(V_y - V_{ter})$$

Separated Form:

$$m dv_y = -b(V_y - V_{ter}) dt$$

$$\frac{m dv_y}{V_y - V_{ter}} = -b dt$$

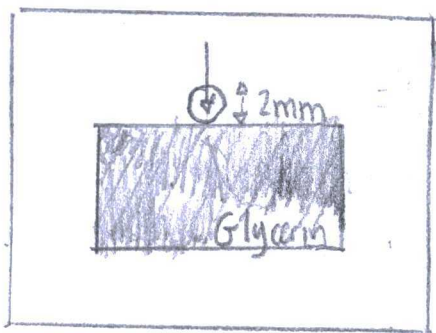
Form integrated:

$$m \int \frac{dv_y}{V_y - V_{ter}} = -b \int_0^t dt$$

$$V_y = V_{ter} + (V_{y0} - V_{ter})e^{-bt}$$

↗ An equation comparable to 2.30, except $b = 1/\tau$.

2.10



"Steel ball bearing dropped into glycerin"

(2.82) "Linear Drag for a sphere"

$$F_{lin} = 3\pi\eta Dv \quad \text{where } \eta = \text{"Viscosity"}$$

$D = \text{"Diameter"}$
 $v = \text{"speed"}$

a) Characteristic time:

(2.25) "Vertical motion with linear drag"

$$\begin{aligned} m\dot{v}_y &= mg - bv_y \\ &= mg - (b_0 + b_{gly})v_y \\ &= m \frac{dv}{dt} \\ dt &= \frac{m dv}{m - (b_0 + b_{gly})v_{lin}} \end{aligned}$$

$$\int_0^t dt = \int_0^{V_{ter}} \frac{m dv}{mg - (b_0 + b_{gly}) V_y}$$

$$t = \frac{m \ln(mg - (b_0 + b_{gly}) V_{ter})}{-(b_0 + b_{gly})}$$

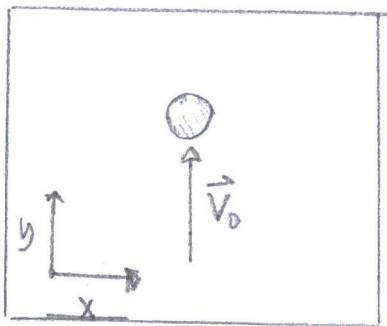
Terminal Speed (V_{ter}):

$$V_{ter} = \frac{-(b_0 + b_{gly}) \cdot t}{m \ln(mg - (b_0 + b_{gly}) V_{ter})}$$

95% Terminal Speed:

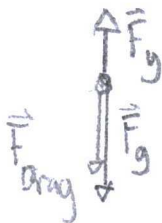
$$t_{95\%} = - \frac{m \ln(mg - 0.95(b_0 + b_{gly}) V_{ter})}{(b_0 + b_{gly})}$$

2.11



"an object...
thrown vertically
up"

a) Free-body Diagram:



Velocity and position:

$$\begin{aligned} \text{y-velocity: } \vec{F} &= m\vec{a} \\ &= m \frac{d\vec{v}}{dt} \end{aligned}$$

$$\int dv = \int a_y dt$$

$$v_y = a_y t + C$$

$$\text{@ } t=0, v=v_0, \text{ so } C=v_0$$

$$v_y = -g \cdot t + v_0$$

$$\begin{aligned} \text{y-position: } v_y &= -g \cdot t + v_0 \\ &= \frac{dy}{dt} \end{aligned}$$

$$\int dy = \int (-g t + v_0) dt$$

$$= -\frac{1}{2} g t^2 + v_0 t + C$$

$$y = -\frac{1}{2} g t^2 + v_0 t + C$$

a) $t=0, y=y_0, \text{ so } C=y_0$

$$y(t) = -\frac{1}{2}gt^2 + v_{y,0}t + y_0$$

b) Maximum Height:

$$\frac{dy(t)}{dt} = \frac{d}{dt} \left(-\frac{1}{2}gt^2 + v_{y,0}t + y_0 \right)$$

$$= -gt + v_{y,0}$$

$$= 0$$

$$t_{\max} = \frac{v_{y,0}}{g}$$

$$y(t)_{\max} = -\frac{1}{2}g\left(\frac{v_{y,0}}{g}\right)^2 + v_{y,0} \frac{v_{y,0}}{g} + y_0$$

$$= +\frac{1}{2} \frac{v_{y,0}^2}{g} + y_0$$

c) Part b assumed no drag coefficient because no statements about shape, medium, or drag.

IF drag, then

(2.13) "Newton's 2nd Law"

$$m\ddot{y} = mg - bV_y$$

= 0 at maximum height

$$\int \frac{dy}{V_y} = \frac{-b}{m} \int dt$$

$$V_y = e^{-b/mt + C}$$

$$= A e^{-b/m \cdot t}$$

$$= \frac{dy}{dt}$$

$$y(t) = A \int_0^t e^{-b/m \cdot t} dt$$

$$= A(1 - e^{-b/m \cdot t})$$

@ $t=0$, $y(t=0) = \frac{b}{m} V_0$, so $A = \frac{b}{m} V_0$

$$y(t) = \frac{m}{b} V_0 (1 - e^{-b/m \cdot t})$$

Drag coefficient approaches Zero:

The maximum height, the position

$$\text{matches } y_{\max}(t_{\max} = \frac{V_{y,0}}{g}) = mg(1 - e^{-b/m \cdot t})$$

$$= \frac{1}{2} \frac{V_{y,0}^2}{g}$$

for $y_0=0$, $b=0$

2.12

V as a function of X:

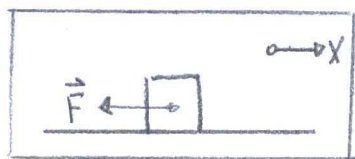
$$\begin{aligned}
 F_{\text{avg}} &= m a_{\text{avg}} \\
 &= m \frac{dv}{dt} \\
 &= m \frac{dx}{dt} \cdot \frac{dv}{dt} \cdot \frac{dt}{dx} \\
 &= m v \cdot \frac{dv}{dx}
 \end{aligned}$$

$$= \frac{m}{2} \frac{d(v^2)}{dx}$$

$$\int_{v_0}^v d(v^2) = \frac{2}{m} \int F(x) dx$$

$$v^2 = v_0^2 + \frac{2}{m} \int F(x) dx$$

2.13



"mass... constrained
on x-axis and
subject to a
net force"

(2.85) "Velocity as a function of x"

$$v^2 = v_0^2 + \frac{2}{m} \int F(x) dx$$

$$= v_0^2 - \frac{2k}{m} \int_0^x x dx$$

$$= v_0^2 - \frac{2k}{m} x$$

$$V = \sqrt{v_0^2 - \frac{2k}{m} x}$$

speed as a
function of
x

X as a function of t:

$$V = \sqrt{V_0^2 - \frac{2k}{m} X}$$

$$= \frac{dX}{dt}$$

$$\int_0^t dt = \int_0^X \frac{1}{\sqrt{V_0^2 - \frac{2k}{m} X}} dX$$

u-substitution on

$$u = V_0^2 - \frac{2k}{m} X$$

$$\frac{du}{dX} = -\frac{2k}{m}$$

$$t = + \frac{m}{k} \sqrt{V_0^2 - \frac{2k}{m} X}$$

$$X = \frac{m}{2k} \sqrt{V_0^2 - \left(\frac{k}{m} t\right)^2}$$

2.14

(From problem) "Force along X-axis"

$$F(v) = -F_0 e^{v/v_0}$$

a) Velocity as a function of time:

(Problem 2.7) "t as a function of v"

$$t = m \int_{v_0}^v \frac{dv'}{F(v')}$$

$$= m \int_{v_0}^v \frac{dv'}{(-F_0 e^{v'/v_0})}$$

$$= -\frac{mV}{F_0} \left[e^{v/v_0} - e^{v_0/v_0} \right] + C$$

$$v' = V_0 \ln \left(\frac{mV}{mV e^{-v_0/V} - F_0 t} \right)$$

b) Time when $v=0$:

$$t = -\frac{mV}{F_0} \left[e^{-v/V} - e^{-v_0/V} \right]$$

$$= -\frac{mV}{F_0} \left[e^0 - e^{-v_0/V} \right]$$

$$= -\frac{mV}{F_0} \left[1 - e^{-v_0/V} \right]$$

An interpretation evaluates the max or min with $v(t)$.

Results were inconclusive.

c) Position:

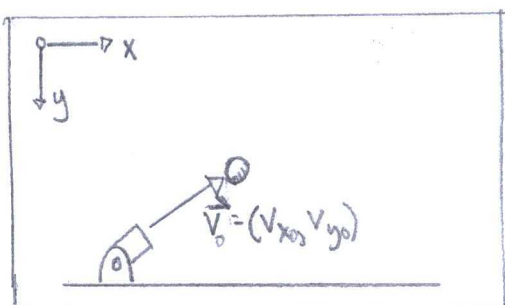
$$x = \int_0^t v(t) dt$$

$$= V \int_0^t \ln \frac{mV}{mVe^{-v_0/V} - F_0 \cdot t} dt$$

$$= V \left[\left[t - \frac{Vm e^{-v_0/V}}{F_0} \right] \ln \left(\frac{mV}{mVe^{-v_0/V} - F_0 t} \right) + t \right]$$

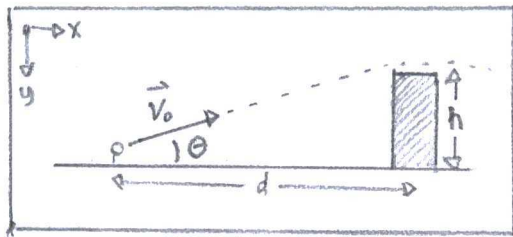
$$- \frac{Vm e^{-v_0/V}}{F_0} \frac{v_0/V}{t}$$

2.15



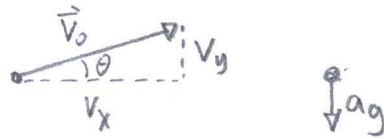
"A projectile launched with a velocity, no air resistance"

2.16.



"What is the minimum speed for which the ball will clear a wall?"

Free-body Diagram:



X-Components:

$$X(t) = X_0 + V_{x0}t + \frac{1}{2}a_x t^2$$

$$= V_0 \cos \theta \cdot t$$

y-components:

$$y(t) = y_0 + V_{y0}t + \frac{1}{2}a_y t^2$$

$$= V_0 \sin \theta t - \frac{1}{2}a_g t^2$$

Minimum Velocity:

$$y(t) = h$$

$$= V_0 \sin \theta t - \frac{1}{2}a_g t^2$$

$$x(t) = d$$

$$= V_0 \cos \theta \cdot t$$

$$t = d / V_0 \cos \theta$$

$$h = V_0 \sin \theta \cdot \frac{d}{V_0 \cos \theta} - \frac{1}{2}a_g \left(\frac{d}{V_0 \cos \theta} \right)^2$$

$$V_0 = \sqrt{\frac{a_g}{2(d \tan \theta - h)}} \cdot \frac{d}{\cos \theta}$$

X-components:

$$x(t) = x_0 + v_{x0} t + \frac{a_x}{2} t^2$$

$$= v_{x0} t$$

y-components:

$$y(t) = y_{x0} + v_{y0} t - \frac{a_y}{2} t^2$$

$$= v_{y0} t - \frac{a_g}{2} t^2$$

Time in air:

$$\frac{dy(t)}{dt} = \frac{d}{dt} \left(v_{y0} t - \frac{a_g}{2} t^2 \right)$$

$$= v_{y0} - a_g t$$

$$= 0$$

$t = v_{y0}/a_g$ is half the time in air.

Distance travelled:

$$x(t = 2v_{y0}/a_g) = \frac{2v_{x0} \cdot v_{y0}}{a_g}$$

Problems with
 $\tan \theta < h/d$

Trigonometric Identity: $\tan \theta = \frac{\sin \theta}{\cos \theta}$

The initial velocity becomes imaginary from $\tan \theta < h/d$. A real solution is more applicable to a golfer.

Initial Conditions:

$$\theta = 25^\circ, d = 50\text{m}, h = 2\text{m}$$

$$V_0 = \sqrt{\frac{4.9\text{m/s}^2}{2(50\text{m}\tan(25^\circ) - 2\text{m})}} \cdot \frac{50\text{m}}{\cos(25^\circ)}$$
$$= 26\text{m/s}$$

2.17

(2.36) "Horizontal and Vertical Range
with Terminal Velocity"

$$x(t) = V_{x0} \tau (1 - e^{-t/\tau})$$

$$y(t) = (V_{y0} + V_{\text{ter}}) \tau (1 - e^{-t/\tau}) - V_{\text{ter}} \cdot t$$

where τ = "Characteristic
Time"
 $= V_{\text{ter}} / g$

y as a function of X:

$$X(t) = V_{x0} \tau (1 - e^{-t/\tau})$$

$$t = \tau \ln \left(1 - \frac{X}{V_{x0} \cdot \tau} \right)$$

$$\begin{aligned} y(t) &= (V_{y0} + V_{ter}) \tau (1 - e^{-t/\tau}) - V_{ter} \cdot t \\ &= \frac{V_{y0} + V_{ter}}{V_{x0}} X + V_{ter} \cdot \tau \ln \left(1 - \frac{X}{V_{x0} \cdot \tau} \right) \end{aligned}$$

2.18

(2.87) "Taylor's Theorem"

$$f(x+\delta) = f(x) + f'(x)\delta + \frac{1}{2!} f''(x)\delta^2 + \frac{1}{3!} f'''(x)\delta^3$$

a) $\ln(1+\delta)$:

$$f(x) = \ln(1+\delta)$$

$$f'(x) = \frac{1}{1+\delta}$$

$$f''(x) = \frac{-1}{(1+\delta)^2}$$

$$f'''(x) = \frac{2}{(1+\delta)^3}$$

Note: Funky
notation
because
 $f(x) \neq f(\delta)$

$$f(x+\delta) = \ln(1+\delta) + \frac{\delta}{1+\delta} - \frac{\delta^2}{2(1+\delta)^2} + \frac{2\delta^3}{6(1+\delta)^3}$$

b) $\cos \delta$:

$$f(x) = \cos \delta$$

$$f'(x) = -\sin \delta$$

$$f''(x) = -\cos \delta$$

$$f'''(x) = \sin \delta$$

$$f(x+\delta) = \cos(\delta) - \sin(\delta)\delta - \frac{1}{2}\cos(\delta)\delta^2 + \frac{1}{6}\sin(\delta)\delta^3$$

c) $\sin(\delta)$:

$$f(x) = \sin \delta$$

$$f'(x) = \cos \delta$$

$$f''(x) = -\sin \delta$$

$$f'''(x) = -\cos \delta$$

$$f(x+\delta) = \sin(\delta) + \cos(\delta)\delta - \frac{1}{2}\sin(\delta)\delta^2 - \frac{1}{6}\cos(\delta)\delta^3$$

d) e^δ :

$$f(x) = e^\delta$$

$$f'(x) = e^\delta$$

$$f''(x) = e^\delta$$

$$F'''(x) = e^{\delta}$$

$$F(x+\delta) = e^{\delta} \left(1 + \delta + \frac{\delta^2}{2!} + \frac{\delta^3}{3!} \right)$$

2.19

a) (2.36) "Horizontal and Vertical Range
with Terminal Velocity"

$$X(t) = V_{x0} \tau (1 - e^{-t/\tau})$$

$$y(t) = (V_{y0} + V_{ter}) \tau (1 - e^{-t/\tau}) - V_{ter} \cdot \tau$$

Without air resistance, the
equations dramatically change,

$$X(t) = V_{x0} t$$

$$y(t) = V_{y0} t - \frac{1}{2} a_g t^2$$

$$y\left(t = \frac{X}{V_{x0}}\right) = \frac{V_{y0}}{V_{x0}} X - \frac{1}{2} \frac{a_g}{V_{x0}^2} X^2$$

b) (2.37) "Vertical range"

$$y(t) = \frac{V_{y0} + V_{ter}}{V_{x0}} X + V_{ter} \cdot \tau \cdot \ln \left(1 - \frac{X}{V_{x0} \tau} \right)$$

$$\lim_{\tau, V \rightarrow \infty} y(t) = \lim_{\tau, V \rightarrow \infty} \left[\frac{V_{y0} + V_{ter}}{V_{x0}} X + V_{ter} \cdot \tau \ln \left(1 - \frac{X}{V_{x0} \tau} \right) \right]$$

Taylor series:

Logarithm

$$\ln(1-\epsilon) = -\epsilon + \frac{\epsilon^2}{2} - \frac{\epsilon^3}{3} + \dots$$

$$\lim_{\tau, v \rightarrow \infty} y(t) = \lim_{\tau, v \rightarrow \infty} \left[\frac{v_{y0} + v_{ter}}{v_{x0}} X - v_{ter} \tau \left(\frac{X}{v_{x0}\tau} + \frac{1}{2} \left(\frac{X}{v_{x0}\tau} \right)^2 + \frac{1}{3} \left(\frac{X}{v_{x0}\tau} \right)^3 + \dots \right) \right]$$

$$= \frac{v_{y0}}{v_{x0}} X + \frac{v_{ter}}{v_{x0}} X - \frac{v_{ter}}{v_{x0}} X - \frac{v_{ter}}{2v_{x0}^2\tau} X^2 - \dots$$

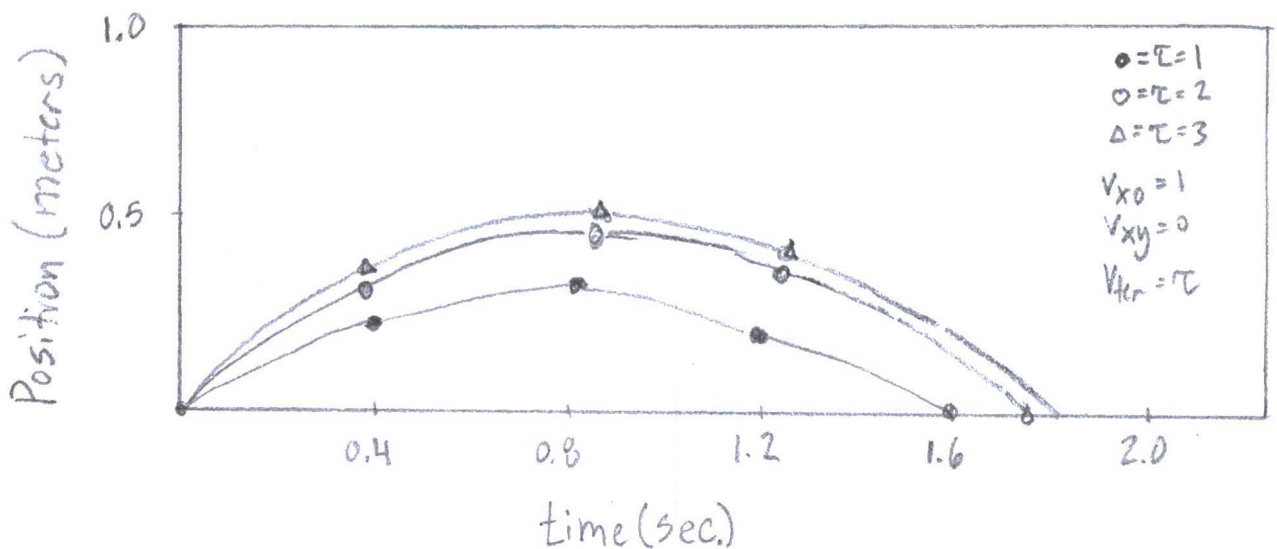
$$= \frac{v_{y0}}{v_{x0}} X - \frac{1}{2} \frac{a_0}{v_{x0}^2} X^2$$

2.20

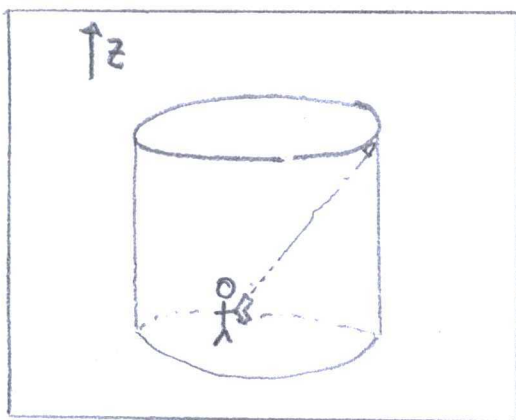
(2.36) "Trajectory and Range in Linear Medium"

$$x(t) = v_{x0} \tau (1 - e^{-t/\tau})$$

$$y(t) = (v_{y0} + v_{ter}) \tau (1 - e^{-t/\tau}) - v_{ter} t$$



2.21

Free-body Diagram:

"Gun... fire shells..
using cylindrical
coordinates"

 ρ -component:

(1.37) "Polar Coord-modified"

$$\rho = \sqrt{x^2 + y^2}$$

$$= v_0 (\cos \theta) t$$

 ϕ -component:

(1.37) "Polar Coordinates-modified"

$$\phi = \arctan(y/x)$$

 z -component:

$$z = z_0 + v_z t - \frac{1}{2} a_z t^2$$

$$= v_0 \sin \theta \cdot t - \frac{1}{2} g t^2$$

$$= \rho \frac{\sin \theta}{\cos \theta} - \frac{g \rho^2}{2 v_0^2 \cos^3 \theta}$$



from ρ -component
 $v_{0\rho} = \rho_0 / \cos \theta$

Sad Day

Author described
Cartesian and
Polar, but not
cylindrical coord.
Well in both
Chapters 1/2.



since
 $t = \frac{\rho_0}{v_0 \cos \theta}$

At maximum,

$$\frac{dz}{d\theta} = \frac{d}{d\theta} \left[p_0 \tan\theta - \frac{g p_0^2}{2V_0^2 \cos^2\theta} \right]$$

$$= p_0 \sec^2\theta - \frac{g \sec^2\theta \tan\theta}{2V_0^2} \cdot p_0^2$$

$$= 0$$

$$\tan\theta = \frac{2 \cdot V_0^2}{p_0 \cdot g}$$

Calculus

Formula needed:

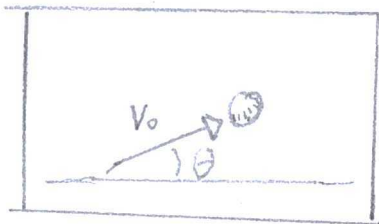
- Quotient Rule

- Chain Rule

$$Z = \frac{2V_0^2}{g} - \frac{g}{2V_0^2} p_0^2 \quad @ 0 \leq \theta \leq \pi/2$$

2.22

(2.39) "Horizontal Range"



"Projectile launched at an angle θ above horizontal ground"

$$\frac{V_{y0} + V_{ter}}{V_{x0}} R + V_{ter} T \ln\left(1 - \frac{R}{V_{x0} T}\right) = 0$$

a) Maximum Range:

$$\text{If } V_{ter} = V_0 = g = 1$$

$$\text{and } \theta = \pi/4$$

then From (2.38)

$$R_{vac} = \frac{2V_{x0}V_{y0}}{g} = \frac{2}{g}$$

b) By a plot, $R = 0, 0.797 \cdot \tau$

c) $\theta = 0.4$, $R = 0.56\text{m}$, $R_{\text{vac}} = 0.71\text{m}$

$\theta = 0.5$, $R = 0.50\text{m}$, $R_{\text{vac}} = 0.94\text{m}$

$\theta = 0.6$, $R = 0.52\text{m}$, $R_{\text{vac}} = 0.93\text{m}$

$\theta = 0.7$, $R = 0.51\text{m}$, $R_{\text{vac}} = 0.96\text{m}$

$\theta = 0.8$, $R = 0.49\text{m}$, $R_{\text{vac}} = 1.00\text{m}$

d) $\theta = 0.01$, $R = 0.02$, $R_{\text{vac}} = 0.02\text{m}$

R_{vac} is always more than R

from drag in medium. A

ball travels farther in a

Vacuum.

2.23

Terminal Air Speeds:

a) Steel ball bearing ($d = 3\text{mm}$):

(2.3) "Quadratic Air resistance"

$$F_{\text{quad}} = C v^2$$

(2.4) "Coefficients"

$$C = \gamma D^2$$

γ = "Medium constant"
 $= 0.25 \text{ N} \cdot \text{s}^2 / \text{m}^4$

D = Diameter

$$F_{\text{quad}} = \gamma D^2 v^2$$

$$v = \sqrt{F_{\text{quad}} / \gamma D^2}$$

$$= \sqrt{m \cdot a / \gamma D^2}$$

$$= \sqrt{\frac{\frac{4}{3} \pi \left(\frac{3 \text{ mm} \times \frac{1 \text{ cm}}{10 \text{ mm}} \right)^3 \cdot 8 \text{ g/cm}^3 \cdot \frac{1 \text{ kg}}{10^3 \text{ g}} \cdot 9.8 \text{ m/s}^2}{0.25 \text{ N} \cdot \text{s}^2/\text{m}^4 \cdot \left(3 \text{ mm} \times \frac{1 \text{ m}}{10^3 \text{ mm}} \right)^2}}$$

$$\approx 2.0 \text{ m/s}$$

b) Steel shot (16-lbs):

$$v = \sqrt{F_{\text{quad}} / \gamma D^2}$$

$$= \sqrt{m \cdot a / \gamma D^2}$$

$$= \sqrt{\frac{16 \text{ lbs} \times \frac{1 \text{ kg}}{2.2 \text{ lbs}} \cdot 9.8 \text{ m/s}^2}{0.25 \text{ N} \cdot \text{s}^2/\text{m}^4 \left(2 \left(\frac{3}{4} \pi \frac{16 \text{ lbs} \times 1 \text{ kg} / 2.2 \text{ lbs}}{8 \text{ g/cm}^3 \cdot \frac{1 \text{ kg}}{10^3 \text{ g}} \left(\frac{10^2 \text{ cm}}{1 \text{ m}} \right)^3 \right)^{1/3} \right)^2}}$$

$$\approx 1000 \text{ m/s}$$

c) Parachutist (200-lbs):

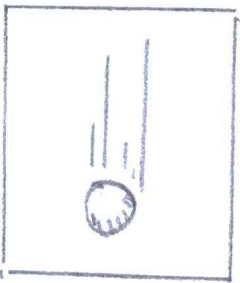
$$v = \sqrt{F_{\text{quad}} / \gamma D^2}$$

$$= \sqrt{m \cdot a / \gamma D^2}$$

$$= \sqrt{\frac{200 \text{ lbs} \times \frac{1 \text{ kg}}{2.2 \text{ lbs}} \cdot 9.8 \text{ m/s}^2}{0.25 \text{ N} \cdot \text{s}^2/\text{m}^4 \left(2 \left(\frac{4}{3} \pi \cdot \frac{1 \text{ g}}{\text{cm}^3} \times \frac{1 \text{ kg}}{10^3 \text{ g}} \cdot \left(\frac{10^2 \text{ cm}}{1 \text{ m}} \right)^3 \right)^{1/3} \right)}}$$

$$\approx 100 \text{ m/s}$$

2.24



"Sphere falling through air"

a) (2.84) "Quadratic Drag"

$$F_{\text{quad}} = K \rho A v^2$$

K = "Number less than one"

ρ = "Density of fluid"

A = "Cross-section"

v = "Velocity"

$$= m \cdot g$$

$$m \cdot g = K \rho A v^2$$

$$v = \sqrt{\frac{4 m g}{\rho \cdot A}}$$

∞ From problem

$$K = 1/4$$

$$= \sqrt{\frac{16 m \cdot g}{\rho_{\text{air}} D^2 \cdot \pi}}$$

$$= \sqrt{\frac{16 V \cdot \rho_{\text{sph}} \cdot g}{\rho_{\text{air}} \cdot D^2 \cdot \pi}}$$

$$= \sqrt{\frac{8}{3} D \cdot g \cdot \left(\frac{\rho_{\text{sph}}}{\rho_{\text{air}}}\right)}$$

Useful

Formula:

$$\text{Diameter} = 2r$$

$$\text{Area} = \pi r^2$$

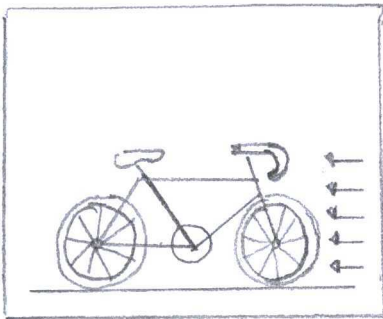
$$\text{Volume} = \frac{4}{3} \pi r^3$$

$$\text{Density} = m/V$$

b) If $\rho_{\text{Heavy}} > \rho_{\text{light}}$, then $V_{\text{Heavy}} > V_{\text{light}}$

c) If $D_{\text{large}} > D_{\text{small}}$, then $V_{\text{large}} > V_{\text{small}}$.

2.25



"Cyclist... halt
under influence
of a quadratic
drag"

Equation 2.49 derived:

$$F = m \cdot a \quad \text{"Newton's 2nd law"}$$

$$= m \frac{dv}{dt}$$

$$= -c v^2 \quad \text{"Quadratic drag"}$$

$$m \int_{v_0}^v \frac{1}{v^2} dv = -c \int_0^t dt$$

$$m \left(\frac{1}{v_0} - \frac{1}{v} \right) = -c t$$

$$V(t) = \frac{V_0}{1 + cV_0 t/m} \quad \text{"Equation 2.49"}$$

$$= \frac{V_0}{1 + t/\tau} \quad \text{...When } \tau = \frac{m}{cV_0}$$

Equation 2.51 derived:

$$V(t) = \frac{V_0}{1 + t/\tau}$$

$$= \frac{dx}{dt}$$

$$\int_{x_0}^x dx = V_0 \int_0^t \frac{dt}{1 + t/\tau}$$

$$x(t) = x_0 + V_0 \tau \ln(1 + t/\tau)$$

↑ "Equation 2.51"

2.26

Characteristic time:

(2.50) "Characteristic time for quadratic drag"

$$\tau = \frac{m}{cV_0}$$

where $m = \text{"mass"}$
 $= 80 \text{ kg}$

$$c = \text{"Drag coefficient"} \\ = 0.20 \text{ N/(m/s)}^2$$

$$V_0 = \text{"Initial velocity"} \\ = 20 \text{ m/s}$$

$$\tau = \frac{80 \text{ Kg}}{0.20 \text{ N/(m/s)}^2 \cdot 20 \text{ m/s}} \\ = 20 \text{ s}$$

Velocity Predicting:

(2.49) "Velocity from quadratic drag"

$$V(t) = \frac{V_0}{1 + t/\tau}$$

$$15 \text{ m/s} : \quad t = \tau \left(\frac{V_0}{V} - 1 \right) \\ = 20 \text{ s} \left(\frac{20 \text{ m/s}}{15 \text{ m/s}} - 1 \right) \\ \approx 7 \text{ sec}$$

$$10 \text{ m/s} : \quad t = \tau \left(\frac{V_0}{V} - 1 \right)$$

$$= 20s \left(\frac{20m/s}{10m/s} - 1 \right)$$

$$\approx 20s$$

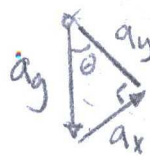
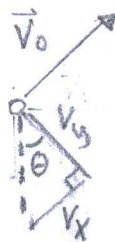
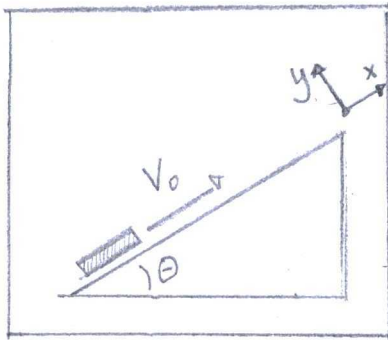
$$5m/s: \quad t = \tau \left(\frac{V_0}{V} - 1 \right)$$

$$= 20s \left(\frac{20m/s}{5m/s} - 1 \right)$$

$$= 60s$$

2.27

Free body Diagram:



X-components:

$$V_x = V_{x,0} + a_{x,0} t$$

$$= V_0 + a_g \sin \theta$$

Y-component:

$$V_y = V_{y,0} + a_{y,0} t$$

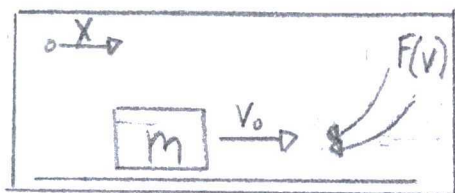
Time upward ($v=0$):

$$t = \frac{m}{c v_{\text{ter}}^2 \sin \theta} \cdot \arctan \left(v_0 / v_{\text{ter}} \sqrt{\sin \theta} \right)$$

2.28

(2.86) "v dv/dx rule"

$$\dot{v} = v \frac{dv}{dx} = \frac{1}{2} \frac{dv^2}{dx}$$



"A mass..coasts
along x axis"

Equation of Motion:

(from problem) "Drag force"

$$F(v) = -c v^{3/2}$$

$$m \cdot \dot{v} = m \cdot v \cdot \frac{dv}{dx}$$

$$= m \cdot v \cdot \frac{dv}{dx}$$

$$dx = \frac{m v dv}{F(v)}$$

$$\int_0^x dx = \int_{v_0}^v \frac{m v dv}{F(v)}$$

$$= a_g \cos \theta \cdot t$$

Velocity with Quadratic Drag:

$$F = m \cdot a$$

$$= m \cdot \frac{dv}{dt}$$

$$= -(m \cdot a_0 + cv^2)$$

$$= -(m \cdot a_g \sin \theta + cv^2)$$

(2.53) "Terminal Velocity"

$$V_{ter} = \sqrt{\frac{mg}{c}}$$

$$m \cdot \frac{dv}{dt} = -c (V_{ter}^2 \sin \theta + v^2)$$

$$\int_{V_0}^v \frac{dv}{(V_{ter}^2 \sin \theta + v^2)} = \frac{-c}{m} \int_0^t dt$$

Integral

$$\int \frac{dx}{x^2 + 1} = \arctan(x) + C$$

$$\frac{1}{V_{ter}^2 \sin \theta} \left[\arctan(V/V_{ter} \sqrt{\sin \theta}) - \arctan(V_0/V_{ter} \sqrt{\sin \theta}) \right] = \frac{-c}{m} t$$

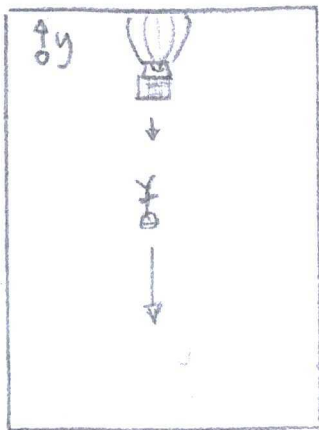
$$V = V_{ter} \sqrt{\sin \theta} \cdot \tan \left(-\frac{c V_{ter}^2 \sin \theta}{m} t + \arctan(V_0/V_{ter} \sqrt{\sin \theta}) \right)$$

$$X = \int_{v_0}^v \frac{m v dv}{-c v^{3/2}}$$

$$= \frac{2m}{c} (\sqrt{v_0} - \sqrt{v})$$

@ $V=0$, $X = 2m\sqrt{v_0}/c$

2.29



"Skydiver₀₀₀
jumps from
a stationary
balloon"

Velocity at a time:

(2.57) "Vertical motion with
Quadratic Drag"

$$V = V_{ter} \cdot \tanh\left(\frac{gt}{V_{ter}}\right)$$

(1.17) "Newton's Second Law
- Modified"

$$a = \frac{F}{m}$$

$$a = \ddot{y}$$

$$\dot{y} = V_0 + \frac{F_0}{m} \cdot t$$

$$m = 70 \text{ kg}, V_{ter} = 50 \text{ m/s}$$

time (s)	1	5	10	20	30
velocity [Drag] (m/s)	0.0	9.7	38	48	50
velocity [No Drag] (m/s)	0.0	9.8	49	98	196

2.30

(from problem 2.23) "Density of a sphere"

$$\text{Density} = 1 \text{ g/cm}^3$$

Mass to volume

$$\text{mass(Kg)} \times \underbrace{\frac{\text{mass(g)}}{\text{mass(Kg)}}}_{\text{mass ratio}} \times \underbrace{\frac{\text{volume(cm}^3\text{)}}{\text{mass(g)}}}_{\text{Density}} = \text{Volume(cm}^3\text{)}$$

$$70 \text{ Kg} \times \frac{1000 \text{ g}}{1 \text{ Kg}} \times \frac{1 \text{ cm}^3}{1 \text{ g}} = 7 \times 10^4 \text{ cm}^3$$

Volume to diameter:

$$\text{Volume(cm}^3\text{)} = \frac{4}{3} \pi (\text{radius(cm)})^3$$

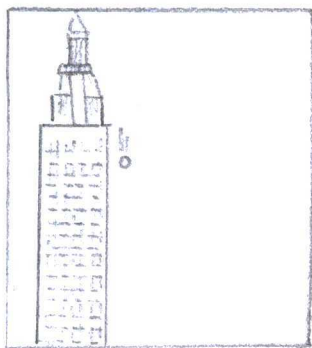
$$\text{radius} = \sqrt[3]{\frac{3}{4\pi} \cdot 7 \times 10^4 \text{ cm}^3}$$

$$= 25 \text{ cm}$$

$$\text{Diameter} = 2 \cdot \text{radius}$$

$$= 50 \text{ cm}$$

2.31



"basketball₀₀₀
dropped from
a 30-m tower"

a) Terminal velocity:

(2.59) "Speed of a baseball"

$$V_{\text{ter}} = \sqrt{\frac{mg}{\gamma D^2}}$$

$$= \sqrt{\frac{600g \times \frac{1\text{kg}}{1000g} \cdot 9.8\text{m/s}^2}{0.25\text{N} \cdot \text{s/m}^4 \cdot (24\text{cm} \times \frac{1\text{m}}{100\text{cm}})^2}}$$

$$\approx 20\text{m/s}$$

b) Time till ground:

(2.58) "y-position"

$$y = \frac{(V_{\text{ter}})^2}{g} \ln \left[\cosh \left(\frac{gt}{V_{\text{ter}}} \right) \right]$$

$$t = \frac{V_{\text{ter}}}{g} \operatorname{arccosh} \left(e^{\frac{y \cdot g / (V_{\text{ter}})^2}{}} \right)$$

$$= \frac{20\text{m/s}}{9.8\text{m/s}^2} \cdot \operatorname{arccosh} \left(e^{\frac{30\text{m} \cdot 9.8\text{m/s}^2}{(20\text{m/s})^2}} \right)$$

$$\approx 2.85$$

How fast?

(2.57) "Velocity with Drag"

$$V = V_{\text{ter}} \cdot \tanh \left(\frac{gt}{V_{\text{ter}}} \right)$$

$$= 20 \text{ m/s} \cdot \tanh\left(\frac{9.8 \text{ m/s}^2 \cdot 2.3 \text{ s}}{20 \text{ m/s}}\right)$$

$$\approx 17 \text{ m/s}$$

If a tower was in a vacuum, then velocity...

$$V = V_0 + g \cdot t$$

$$= 9.8 \text{ m/s} \cdot 2.3 \text{ s}$$

$$= 22.7 \text{ m/s}$$

2.32

a) In the limit to small sizes and low densities, velocities are not terminal velocities during projectile motion.

b) (2.26) "Terminal speed"

$$V_{\text{ter}} = \frac{mg}{b} \quad [\text{for linear drag}]$$

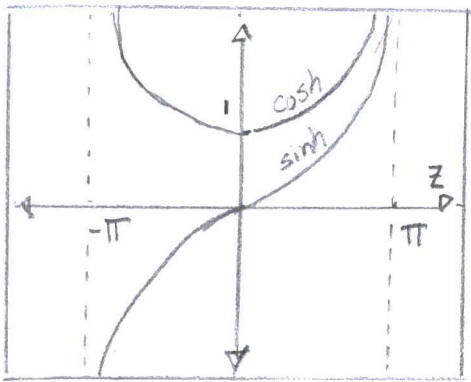
(2.53) "Terminal velocity with Quadratic Drag"

$$V_{\text{ter}} = -\sqrt{\frac{mg}{c}}$$

Quadratic drag depends on density

less by $\sqrt{\text{Density}}$. So in projectile motions, the limit is more applicable to quadratic values.

2.33



a) (from problem) "Hyperbolic Functions"

$$\cosh(z) = \frac{e^z + e^{-z}}{2}$$

$$\sinh(z) = \frac{e^z - e^{-z}}{2}$$

$$b) \cosh(z) = \frac{e^z + e^{-z}}{2}$$

$$= \frac{e^{-i(i)z} + e^{i(i)z}}{2}$$

$$= \cos(iz)$$

$$\sinh(z) = \frac{e^z - e^{-z}}{2}$$

$$= \frac{e^{-i(i)z} - e^{i(i)z}}{2}$$

$$= -i \sin(iz)$$

$$c) \cosh'(z) = \sinh(z)$$

$$\sinh'(z) = \cosh(z)$$

$$d) \int \cosh^2(z) dz = \sinh(z) + C$$

$$\int \sinh(z) dz = \cosh(z) + C$$

$$\begin{aligned} d) \cosh^2(z) - \sinh^2(z) &= \frac{(e^z + e^{-z})}{2} \frac{(e^z + e^{-z})}{2} - \frac{(e^z - e^{-z})}{2} \frac{(e^z - e^{-z})}{2} \\ &= \frac{e^{2z} + 1 + e^{-2z}}{4} - \frac{e^{2z} - 1 - e^{-2z}}{4} \\ &= 4/4 \\ &= 1 \end{aligned}$$

$$\begin{aligned} e) \int \frac{dx}{\sqrt{1+x^2}} &= \int \frac{\cosh(u) du}{\sqrt{1+(\sinh u)^2}} \\ &= \int du \end{aligned}$$

u -substitution $x = \sinh(u)$ $\frac{dx}{du} = \cosh(u)$

$$= u + C$$

$$= \operatorname{arcsinh}(x) + C$$

2.34.

a) (from problem) "Hyperbolic function"

$$\tanh(z) = \sinh(z) / \cosh(z)$$

$$= \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$= \frac{-i \sin(iz)}{\cos(iz)}$$

$$= -i \tan(iz)$$

$$b) \tanh'(z) = \operatorname{sech}^2(z)$$

$$c) \int \tanh(z) dz = \int \frac{\sinh(z)}{\cosh(z)} dz$$

$$= \int \frac{1}{u} du$$

<u>u-substitution</u> $u = \cosh(z)$ $\frac{du}{dz} = \sinh(z)$

$$= \ln(u)$$

$$= \ln(\cosh(z))$$

d) (from problem) "hyperbolic relation"

$$1 - \tanh^2(z) = 1 - \frac{\sinh^2(z)}{\cosh^2(z)}$$

$$= \frac{\cosh^2(z) - \sinh^2(z)}{\cosh^2(z)}$$

$$= \frac{1}{\cosh^2(z)}$$

$$= \operatorname{sech}^2(z)$$

$$e) \int \frac{dx}{(1-x^2)} = \int \frac{\operatorname{sech}^2(u)}{\operatorname{sech}^2(u)} du$$

$$= \int du$$

$$= u + c$$

$$= \operatorname{arctanh}(x) + c$$

u -Sub: $u = \tanh(x)$ $\frac{du}{dx} = \frac{1}{\operatorname{sech}^2(x)}$

2.35

a) (2.52) "Vertical motion with Quadratic Drag"

$$m\dot{v} = mg - cv^2$$

This equation became (2.57) and (2.58) by an assumption, "acceleration stops at terminal velocity." The formula $v_{\text{ter}} = \sqrt{\frac{mg}{c}}$ aided the integrals in later steps.

b) (from problem) "Parameter"

$$\tau = v_{\text{ter}} / g$$

(2.57) "Velocity with drag"

$$v = v_{\text{ter}} \cdot \tanh\left(\frac{gt}{v_{\text{ter}}}\right)$$

$$= V_{\text{ter}} \cdot \tau \cdot \left[\frac{1}{2} \left(\frac{t}{\tau} \right)^2 - \dots \right]$$

$$= \frac{V_{\text{ter}} \cdot t^2}{2\tau} + \text{const}$$

$$\approx V_{\text{ter}} \cdot t + \text{const}$$

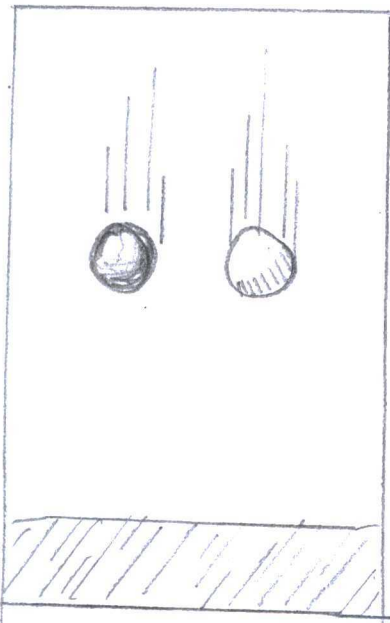
when $t \geq 2\tau$

$$e) y = \frac{(V_{\text{ter}})^2}{g} \ln [\cosh(t/\tau)]$$

$$= \frac{g\tau^2}{2} \left(\frac{t}{\tau} \right)^2 + \dots$$

$$= \frac{1}{2} g t^2$$

2.36



"Aristotle vs.
Galileo"

a) Terminal Speeds:

(2.53) "Terminal velocity"

$$V_{\text{ter}} = \sqrt{\frac{mg}{c}}$$

$$= \sqrt{\frac{m \cdot g}{c \cdot D^2}}$$

$$= \sqrt{\frac{m \cdot g}{\gamma \left(\sqrt[3]{\frac{6m}{\rho}} \right)^2}}$$

$$= v_{ter} \tanh(t/\tau)$$

(2.58) "Position with drag"

$$y = \frac{(v_{ter})^2}{g} \ln \left[\cosh \left(\frac{gt}{v_{ter}} \right) \right]$$

$$= \frac{(v_{ter})^2}{g} \ln \left[\cosh(t/\tau) \right]$$

Percent (time) terminal velocity:

$$@ t = \tau, v = v_{ter} \cdot \tanh(1)$$

$$= v_{ter} \cdot 0.76$$

$$@ t = 2\tau, v = v_{ter} \cdot \tanh(2)$$

$$= v_{ter} \cdot 0.96$$

$$@ t = 3\tau, v = v_{ter} \cdot \tanh(3)$$

$$= v_{ter} \cdot 0.99$$

$$c) @ t \gg \tau, y = \frac{(v_{ter})^2}{g} \ln \left[\cosh \left(\frac{t}{\tau} \right) \right]$$

$$= \frac{(v_{ter})^2}{g} \ln \left[1 + \frac{1}{2} \left(\frac{t}{\tau} \right)^2 + \dots \right]$$

$$100\text{lb-ball}, V_{\text{ter}} = 169.24 \text{ m/s}$$

$$1\text{lb-ball}, V_{\text{ter}} = 78.57 \text{ m/s}$$

b) (2.58) "Position with drag"

$$y = \frac{(V_{\text{ter}})^2}{g} \ln \left[\cosh \left(\frac{gt}{V_{\text{ter}}} \right) \right]$$

Time till ground:

$$t = \frac{V_{\text{ter}}}{g} \operatorname{arccosh} \left(e^{y \cdot g / V_{\text{ter}}^2} \right)$$

$$100\text{lbs ball}, t = 6.46 \text{ seconds}$$

$$1\text{lb ball } t = 6.73 \text{ seconds}$$

$$\Delta t = t_1 - t_{100} = 0.27 \text{ seconds}$$

Both Aristotle and Galileo
were incorrect!

2.37

(2.57) "Velocity with Quadratic Drag"

$$V = V_{\text{ter}} \tanh \left(\frac{gt}{V_{\text{ter}}} \right)$$

(2.55) "Velocity differential"

$$\frac{dv}{1 - v^2/V_{\text{ter}}^2} = g dt$$

Method #1: Hyperbolic Functions

$$\int \frac{du}{(1-u^2)} = \operatorname{arctanh}(u)$$

Method #2: Partial Fractions

$$\int \frac{du}{(1-u^2)} = \frac{1}{2} \int \left(\frac{1}{1+u} + \frac{1}{1-u} \right) du$$

$$\int \frac{dv}{(1-v^2/v_{ter}^2)} = \frac{1}{2} \int \left(\frac{1}{1+v/v_{ter}} + \frac{1}{1-v/v_{ter}} \right) dv$$

$$= \frac{1}{2} \left(v_{ter} \int \frac{1}{k} dk - v_{ter} \int \frac{dg}{g} \right)$$

$$= \frac{v_{ter}}{2} [\ln(k) - \ln(g)] + C$$

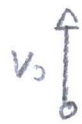
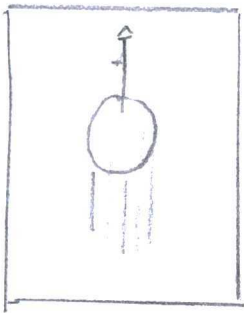
$$= \frac{v_{ter}}{2} \ln \left(\frac{1+v/v_{ter}}{1-v/v_{ter}} \right) + C$$

$$= gt$$

$$@ t=0, v=0, \text{ and } C=0$$

$$v = \frac{e^{2gt/v_{ter}} - 1}{e^{2gt/v_{ter}} + 1}$$

2.38

Free-body Diagram:

"Projectile...
thrown up"

a) Equations of motion:

(2.52) "Vertical motion with
Quadratic Drag"

$$m \dot{v}_y = mg - cv^2$$

$$v_y = v_{ter} \cdot \tanh\left(\frac{gt}{v_{ter}}\right)$$

Assumption at

apex:

$$m \dot{v} = 0$$

b) Top of Trajectory:

$$t = \frac{v_{ter}}{g} \operatorname{arctanh}\left(v_0/v_{ter}\right)$$

c) Baseball values:

$$v_{ter} = 35 \text{ m/s}$$

v_0 (m/s)	t_0 (s)	t in vacuum (s)
1	0.1	0.1
10	1	1
20	2	2
30	5	3
40	5	4

In a vacuum, no terminal velocity exists. So, the ball travels faster without a maximum speed and no drag.

2.39.

a) Equation of motion for a cyclist:

$$m \dot{v} = -cv^2 - f_{fr} - f$$

$$= -cv^2 - 3$$

$$-m \int_{v_0}^0 \frac{dv}{cv^2 + 3} = \int_0^t dt$$

$$-m \int_{v_0}^0 \frac{\sqrt{3} \sec^2(u)}{\sqrt{c}(3 \tan^2(u) + 3)} du = t$$

$$\frac{-m}{\sqrt{3c}} \int_{v_0}^0 du = t$$

$$t = + \frac{m}{\sqrt{3c}} \arctan\left(\frac{\sqrt{c}}{\sqrt{3}} v_0\right)$$

U-substitution:

$$v = \frac{\sqrt{3}}{\sqrt{c}} \tan(u)$$

$$\frac{dv}{du} = \frac{\sqrt{3}}{\sqrt{c}} \sec^2(u)$$

Trigonometric Identity:

$$\tan^2(u) + 1 = \sec^2(u)$$

(from problem) "Drag Force"

$$f = -bv - cv^2$$

Equation of motion:

$$m\dot{v} = +f$$
$$= -bv - cv^2$$

$$m \int \frac{dv}{-bv - cv^2} = \int dt$$

$$-m \int \frac{dv}{v(b+cv)} = t$$

$$-m \left[\frac{1}{b} \int \frac{dv}{v} - \frac{c}{b} \int \frac{dv}{b+cv} \right] = t$$

Partial Fractions

$$\frac{1}{v(b+cv)} = \frac{A}{v} + \frac{B}{b+cv}$$

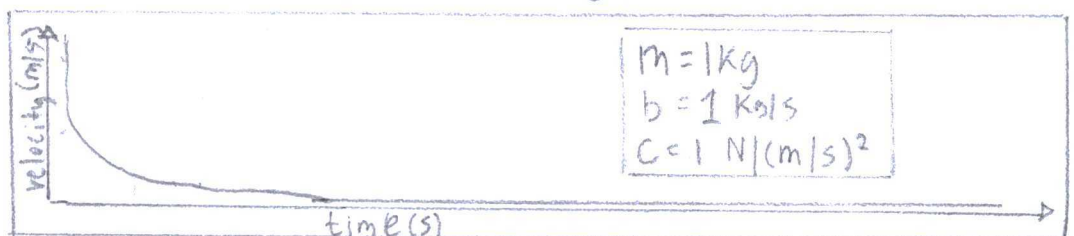
$$1 = A(b+cv) + B(v)$$

$$@ v=0, A = 1/b$$

$$@ v=-b/c, B = -c/b$$

$$-\frac{m}{b} [\ln(v) - \ln(b+cv)] = t$$

$$v = \frac{b}{e^{bt/m} - c}$$



b) Numbers from Problem 2.26:

$$m = 80 \text{ Kg}$$

$$C = 0.20 \text{ N}/(\text{m/s})^2$$

$L = 20 \text{ m}$
If $v_0 = 20 \text{ m/s}$, $t \approx 140 \text{ s}$

If $v_0 = 15 \text{ m/s}$, $t \approx 140 \text{ s}$

↑ "...about the same time with significant figures"

If $v = 10 \text{ m/s}$ and $v_0 = 20 \text{ m/s}$
then $t \approx 19 \text{ seconds}$

If $v = 5 \text{ m/s}$ and $v_0 = 20 \text{ m/s}$
then $t \approx 40 \text{ seconds}$

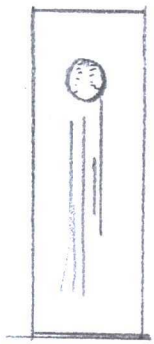
With so many combinations, γ
in a vacuum, t_{max} at $v_0 = 20 \text{ m/s}$
is 2 seconds.

2.40.



"Object coasting horizontally"

2.41

Force body Diagram:Equation of motion:

$$m\dot{v} = -mg - cv^2$$

When $\dot{v} = 0$, then

$$v_{ter} = \sqrt{\frac{mg}{c}}$$

$$m\dot{v} = -mg \text{ and}$$

$$\dot{v} = -g\left(1 + \frac{v^2}{v_{ter}^2}\right)$$

"v dv/dx rule":

(2.86) "v as a function of x - modified"

$$\dot{v} = v \frac{dv}{dy} = \frac{1}{2} \frac{dv^2}{dy}$$

$$= -g\left(1 + \frac{v^2}{v_{ter}^2}\right)$$

$$\int_{v_0}^v \frac{v dv}{\left(1 + \frac{v^2}{v_{ter}^2}\right)} = -g \int_0^y dy$$

Position by
Velocity

$$y = -\frac{v_{ter}^2}{2g} \left[\ln \left(\frac{v^2 + v_{ter}^2}{v_0^2 + v_{ter}^2} \right) \right]$$

Velocity
by
Position

$$V = \sqrt{(V_0^2 + V_{ter}^2) e^{-2gy/V_{ter}^2} - V_{ter}^2}$$

Maximum Height:

$V=0$ at the apex in the throw.

$$y_{max} = \frac{V_{ter}^2}{2g} \ln \left(\frac{V_0^2 + V_{ter}^2}{V_{ter}^2} \right)$$

Parameters from Example 2.5:

$$m = 0.15 \text{ Kg}$$

$$c = \gamma D^2 \quad \text{where } \gamma = 0.25 \text{ N} \cdot \text{s}^2 / \text{m}^4$$

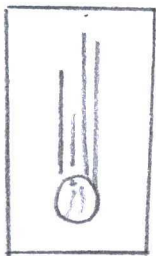
$$D = 7 \text{ cm}$$

$$\begin{aligned} y_{max} &= \frac{V_{ter}^2}{2g} \ln \left(\frac{V_0^2 + V_{ter}^2}{V_{ter}^2} \right) \\ &= \frac{mg/c}{2g} \ln \left(\frac{V_0^2 + mg/c}{mg/c} \right) \end{aligned}$$

$$\approx 17 \text{ m}$$

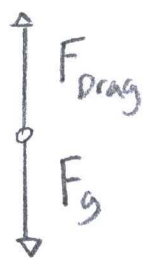
In a vacuum, $y_{max} \approx 20 \text{ m}$.

2.42



"Baseball_{oo}
for the
downward
journey"

Free-body Diagram:



Equation of Motion:

$$m\dot{v} = mg - cv^2$$

When $\dot{v} = 0$, $v_{ter} = \sqrt{\frac{mg}{c}}$

and $\dot{v} = g\left(1 - \frac{v^2}{v_{ter}^2}\right)$

"v dv/dx rule":

(2.36) "v as a function of x-modified"

$$\dot{v} = v \frac{dv}{dy} = \frac{1}{2} \frac{dv^2}{dy}$$

$$= +g\left(1 - \frac{v^2}{v_{ter}^2}\right)$$

$$\int_{v_0}^v \frac{v dv}{(1 - v^2/v_{ter}^2)} = -g \int_{y_{max}}^y dy$$

Note: Book left
initial speed
falling at
Velocity-Zero (v_0)

$$(y_{\max} - y) = \frac{V_{\text{ter}}^2}{2g} \left[\ln \left(\frac{V_{\text{ter}}^2 - V^2}{V_{\text{ter}}^2 - V_0^2} \right) \right]$$

Velocity
by
position

$$V = \sqrt{(V_{\text{ter}}^2 - V_0^2) e^{-2g(y_{\max} - y)/V_{\text{ter}}^2} + V_{\text{ter}}^2}$$

With $y_{\max} = x_{\infty}$ unusual steps

$$= \sqrt{(V_{\text{ter}}^2 - V_0^2) \cdot \left(\frac{V_{\text{ter}}^2}{V_{\text{ter}}^2 + V_0^2} \right) + V_{\text{ter}}^2}$$

$$= \frac{V_{\text{ter}} \cdot V_0}{\sqrt{V_{\text{ter}}^2 + V_0^2}}$$

$$= \frac{\sqrt{\frac{mg}{c}} \cdot V_0}{\sqrt{\frac{mg}{c} + V_0^2}}$$

$$\approx 17 \text{ m/s}$$

Large values of air resistance lower velocity; while low values raise velocity. In a vacuum the speed is 28 m/s.

2.43

a) Height = 2m

Initial velocity = 15 m/s

Angle above horizontal = 45°

(2.61) "Equations of motion"

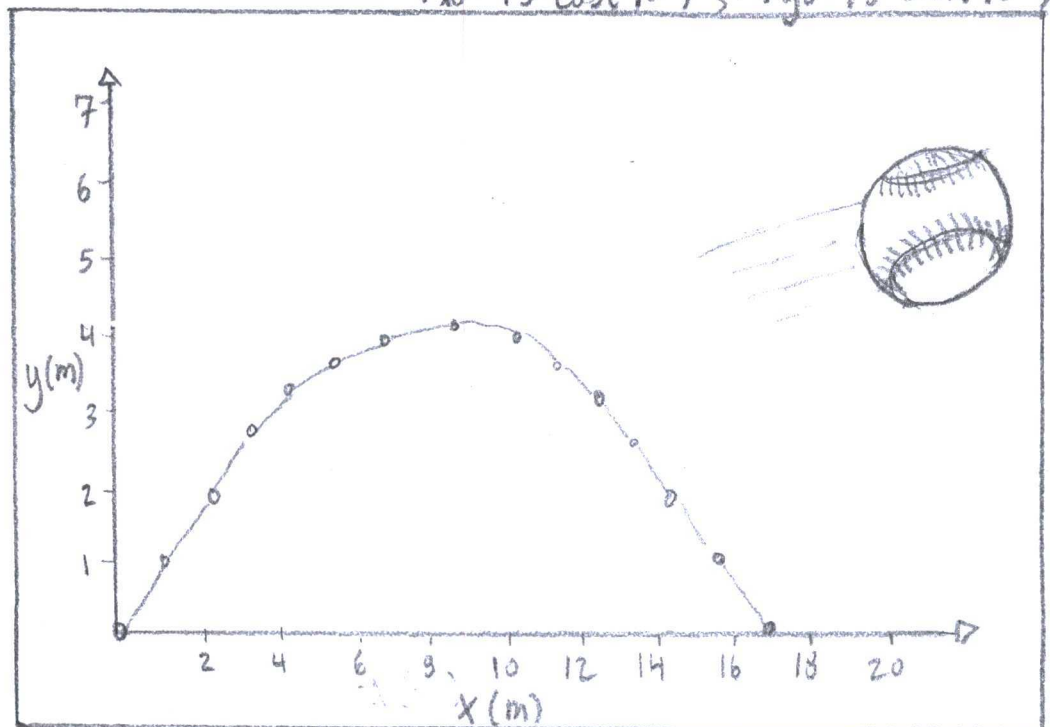
$$m\dot{V}_x = -C\sqrt{V_x^2 + V_y^2} V_x$$

$$m\dot{V}_y = -mg - C\sqrt{V_x^2 + V_y^2} V_y$$

Position:

Runge-Kutta 4th order

Parameters: $h=0.1$, $c=0.001225$, $m=0.15$
 $V_{x0}=15\cos(45^\circ)$, $V_{y0}=15\sin(45^\circ)$



2.44

a) Equations of Motion:

$$m\dot{v}_x = -c(y)v^2$$

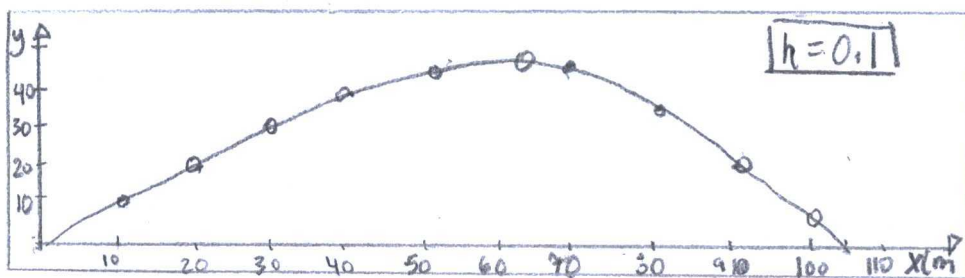
$$= -\gamma D^2 e^{-y/\lambda} \cdot v^2 \quad \text{where } \gamma = 0.25 \text{ N}\cdot\text{s}/\text{m}^4$$

$$D = 15 \text{ cm}$$

$$\lambda \approx 10,000 \text{ m}$$

$$m\dot{v}_y = -mg - cv^2$$

$$= -mg - \gamma D^2 e^{-y/\lambda} \cdot v^2$$

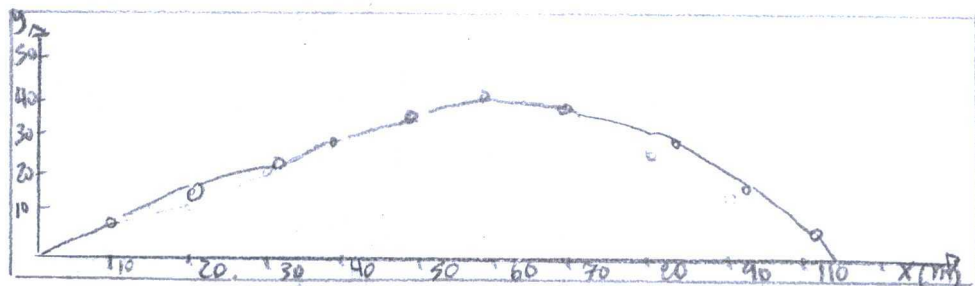
b) Equations of Motion

$$m\dot{v}_x = -cv^2$$

$$= -\gamma D^2 e^{-y/\lambda} \cdot v^2$$

$$m\dot{v}_y = -mg - cv^2$$

$$= -mg - \gamma D^2 e^{-y/\lambda} \cdot v^2$$



$$b) Z = 3 + 4i$$

$$= 5e^{i\theta}$$

$$c) Z = 2e^{-i\pi/3}$$

$$= x + i \cdot x \tan\left(\frac{\pi}{3}\right)$$

$$= 1 + i\sqrt{3}$$

$$Z = \sqrt{x^2 + x^2 \tan^2(\pi/3)}$$

$$x = 1$$

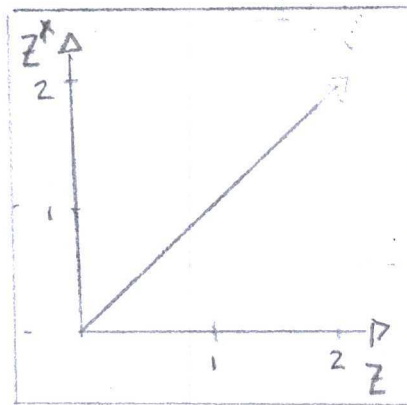
2.46

$$a) \underline{Z = 1 + i} : \text{Real Part : } 1$$

Imaginary Part : 1

Absolute value : $\sqrt{2}$

Phase Angle : $\pi/4$



$$b) \underline{Z = 1 - i\sqrt{3}} : \text{Real part : } 1$$

Imaginary part : $-\sqrt{3}$

Absolute value : 2

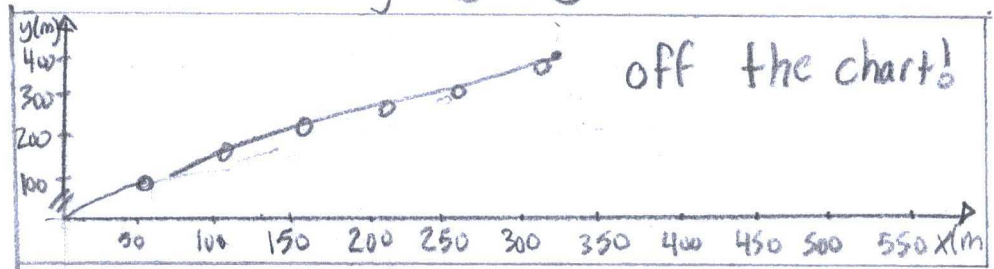
Phase Angle : $\pi/3$

Without air resistance:

Equations of Motion:

$$X = V_{0x} t$$

$$y = V_{0y} t - \frac{1}{2} g t^2$$



2.45

a) (2.76) "Euler's relation"

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$Z = X + iy$$

$$= r \left(\frac{x}{r} + i \frac{y}{r} \right)$$

$$= r (\cos\theta + i\sin\theta)$$

$$= r e^{i\theta}$$

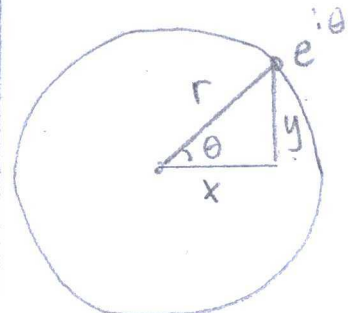
"r" is equally significant
to " θ " in a triangle.

Acronym

Soh Cah Toa

$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$

$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$



2.47.

$$a) \underline{Z = 6 + 0i \quad \text{and} \quad w = 3 - 4i}$$

$$Z + w = 9 - 4i$$

$$Z - w = 3 + 12i$$

$$Z \cdot w = 50$$

$$Z/w = \frac{-14 + 48i}{25}$$

$$b) \underline{Z = 8e^{i\pi/3} \quad \text{and} \quad w = 4e^{i\pi/6}}$$

$$Z + w = (4 + 2\sqrt{3}) + (2 + 4\sqrt{3})i$$

$$Z - w = (4 - 2\sqrt{3}) + (-2 + 4\sqrt{3})i$$

$$Z \cdot w = 32i$$

$$Z/w = \sqrt{3} + i$$

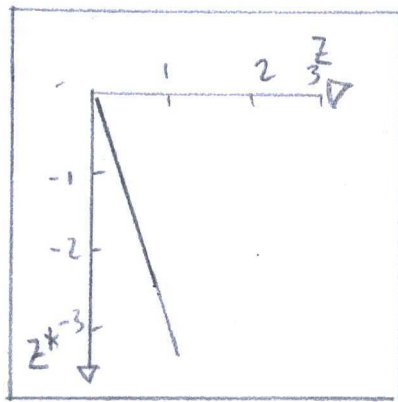
2.48

(From problem) "Absolute complex number"

$$|Z| = \sqrt{x^2 + y^2} \quad (\text{Unusual notation})$$

$$= \sqrt{(x + iy)(x - iy)} \quad \text{"Hermitian product"}$$

$$= \sqrt{Z^* \cdot Z}$$

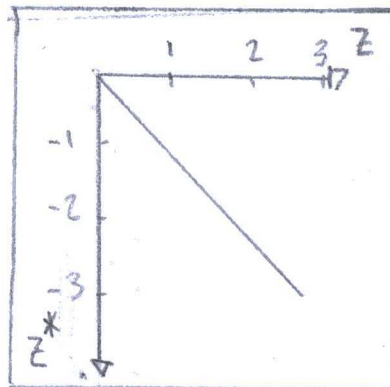


c) $z = \sqrt{2} e^{-i\pi/4}$: Real part : 1

Imaginary Part : -1

Modulus : $\sqrt{2}$

Phase Angle : $-\pi/4$

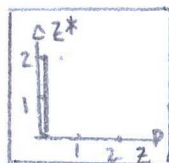


d) $z = 5e^{i\omega t}$: Real part : 0

Imaginary part : 5

Modulus : 5

Phase Angle : 1



2.49

$$(2a) \quad z^2 = (\cos \theta + i \sin \theta)^2$$

$$= \cos^2 \theta + 2 \cos \theta i \sin \theta - \sin^2 \theta$$

$$= \cos 2\theta + i \sin 2\theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

Identity #1

$$2 \sin \theta \cos \theta$$

$$= \sin 2\theta$$

$$z^2 = (\cos \theta + i \sin \theta)^2$$

$$= \cos^2 \theta + 2 \cos \theta i \sin \theta - \sin^2 \theta$$

$$= \cos 2\theta + i \sin 2\theta$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$b) \quad z^3 = (\cos \theta + i \sin \theta)^3$$

$$= \cos^3 \theta - 3 \sin^2 \theta \cos \theta + 3 i \sin \theta \cos^2 \theta - i \sin^3 \theta$$

$$= \cos(3\theta) + i \sin(3\theta)$$

$$\cos(3\theta) = \cos^3 \theta - 3 \sin^2 \theta \cos \theta$$

$$z^3 = (\cos \theta + i \sin \theta)^3$$

$$= \cos^3 \theta - 3 \sin^2 \theta \cos \theta + 3 i \sin \theta \cos^2 \theta - i \sin^3 \theta$$

$$= \cos(3\theta) + i \sin(3\theta)$$

$$\sin(3\theta) = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$$

2.50

(2.72) "Taylor series"

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$$

$$\frac{de^z}{dz} = 0 + 1 + z + \frac{z^2}{2} + \dots$$

$$= e^z$$

2.51

(2.72) "Taylor series"

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$$

$$e^z \cdot e^w = \left(1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots\right) \left(1 + w + \frac{w^2}{2!} + \frac{w^3}{3!} + \dots\right)$$

$$= \frac{1}{1} + \frac{zw}{1} + \frac{z^2 w^2}{2!} + \frac{z^3 w^3}{3!} + \frac{z^2}{2!} + \frac{z^2 w}{2!} + \frac{z^2 w^2}{2!} + \frac{z^2 w^3}{3!} + \frac{z^3}{3!} + \frac{z^3 w}{3!} + \frac{z^3 w^2}{3! 2!} + \frac{z^3 w^3}{3! 3!} + \dots$$

$$= 1 + z + \dots + w + \dots$$

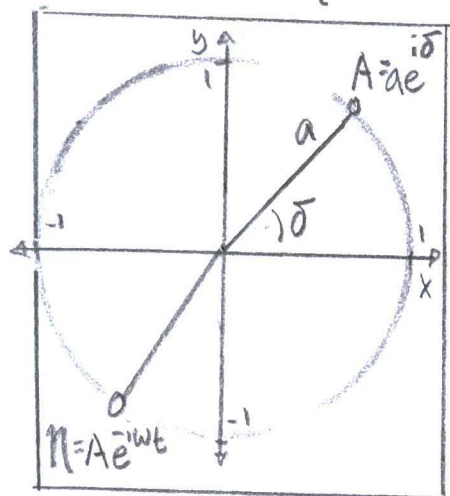
$$= 1 + (z+w) + \left(\frac{z^2}{2!} + \frac{w^2}{2!}\right) + \left(\frac{z^3}{3!} + \frac{w^3}{3!}\right) + \dots$$

$$= e^{z + w}$$

2.52

(from

problem) "Transverse velocity"



"Unit circle"

$$\eta = V_x + i V_y$$

(2.77) "Transverse velocity"

$$\eta = A e^{-i\omega t} = a e^{i\delta} e^{-i\omega t} = a e^{i(\delta - \omega t)}$$

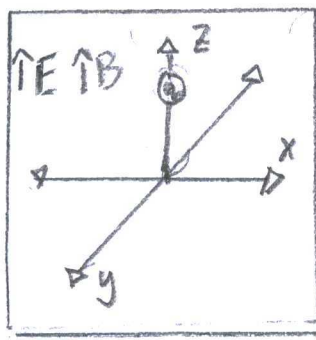
$$= a(\cos(\delta - \omega t) + i \sin(\delta - \omega t))$$

$$V_x = a \cos(\delta - \omega t)$$

$$V_y = a \sin(\delta - \omega t)$$

Both velocities shift to the right with additional time.

2.53.



"Charged particle ... moves in a uniform electric and magnetic fields, $\vec{E}$ and $\vec{B}$ "

(from problem) "Net Force"

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

Equation of Motion:

$$m\dot{\vec{v}} = q(\vec{E} + \vec{v} \times \vec{B})$$

$$= q(\vec{E}_z + \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & v_z \\ 0 & 0 & B_z \end{vmatrix})$$

$$= q(\vec{E}_z + v_y B_z \hat{x} - v_x B_z \hat{y})$$

$$m\dot{v}_x = q v_y B_z \hat{x}$$

$$m\dot{v}_y = -q v_x B_z \hat{y}$$

} "Transverse velocity"

$$m\dot{v}_z = q E_z$$

2.54.

(2.68) "Cyclotron velocity"

$$\dot{v}_x = \omega v_y$$

$$\dot{v}_y = -\omega v_x$$

$$\text{where } \omega = \frac{qB}{m}$$

$$\begin{aligned} \text{a) } \ddot{v}_x &= \omega \dot{v}_y \\ &= -\omega^2 v_x \end{aligned}$$

↑ "Second order linear differential equation"

A general solution: $V_x = \sin(\omega t)$

b) But Form a match to (2.77),

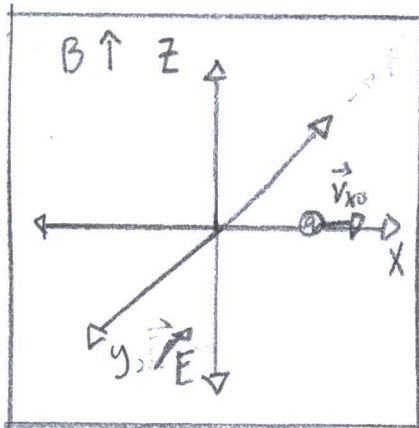
$$V_x = A[\cos(\omega t) + i \sin(\omega t)]$$

$$= A e^{-i\omega t}$$

2.55

a) Equation of Motion:

(problem 2.53) "Lorentz Equation"



$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

$$= q(E_y + \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & v_z \\ 0 & 0 & B_z \end{vmatrix})$$

$$= q(E_y + v_y B_z \hat{x} - v_x B_z \hat{y})$$

$$m \dot{v}_x = q v_y B_z \hat{x}$$

$$m \dot{v}_y = q E_y - v_x B_z \hat{y}$$

$$m \dot{v}_z = 0$$

b) Drift Speed:

$$m \dot{v}_x = v_y B_z \quad \rightarrow \int_0^{v_x} dv$$

$$v_x = \frac{v_y}{m} B_z t + C$$

$$c) \quad = \frac{v_y}{m} B_z t + v_{x0} \quad @ (v_x(t=0) = v_{x0})$$

d) Position:

$$X = X_0 + v_{x0} t + \frac{v_y}{2m} B_z t^2$$